

Sales Performance Dashboard (2023–2025)

1. Project Overview

The **Sales Performance Dashboard (2023–2025)** provides an interactive view of sales performance, customer behavior, regional performance, delivery metrics, and product trends. The project focuses on transforming raw sales data into meaningful insights using **Excel and Power Query**.

2. Dataset Description

The dataset contains sales transaction records covering the period **2023–2025**.

Key fields include:

Date | Region | Product | Quantity | UnitPrice | StoreLocation | CustomerType | Discount | Salesperson | TotalPrice | PaymentMethod | Promotion | Returned | OrderID | CustomerName | ShippingCost | OrderDate | DeliveryDate | RegionManager

The dataset was analyzed after cleaning and validation to ensure reliable dashboard results.

3. Data Cleaning Report

3.1 Objective

The objective of data cleaning was to improve the **accuracy, consistency, and reliability** of the sales data before performing analysis and creating the dashboard.

3.2 Data Cleaning Steps

Step	Cleaning Activity	Action Taken	Reason / Why This Step Was Taken
1	Remove Duplicates	Identified and removed duplicate records.	To avoid counting the same transaction more than once and prevent incorrect sales calculations.
2	Handle Missing Values	Checked blank/null values and handled them without changing the original meaning of the data.	To prevent missing values from affecting analysis while preserving the meaning of the available information.

3	Standardize Text	Corrected inconsistent spelling, capitalization, and spacing.	To ensure the same categories are treated as one value in Pivot Tables and slicers.
4	Date Formatting	Converted dates into a consistent date format.	To enable accurate date-based analysis, timelines, and filtering.
5	Numeric Formatting	Ensured Quantity, Unit Price, Discount, and Sales values were numeric and non-negative.	To ensure accurate calculations and prevent invalid numerical values from affecting KPIs.
6	Validate Values	Checked for invalid or unrealistic values.	To ensure the dataset contains logically valid values before analysis.
7	Outlier Detection	Identified unusually high or low values.	To investigate unusual records and determine whether they were genuine observations or data errors.
8	Data Consistency	Ensured columns followed a consistent structure and data type.	To maintain a standardized dataset and avoid errors during analysis.

3.3 Missing-Value Insight – Promotion Column

The **Promotion** column contained **370 blank values**, representing **approximately 25% of the column**.

Among these 370 records:

- **280 records had discounts**
- **90 records had no discount**

Since the actual promotion status could not be reliably inferred from the discount value alone, the blank promotion values were classified as **"Unknown"** in the cleaned dataset.

Missing-Value Handling Result

Before Cleaning	Analysis	Action Taken	After Cleaning
370 blank Promotion values	280 had discounts and 90 had no discount	Actual promotion status could not be reliably inferred	All 370 blank values classified as "Unknown"

Result: No records were deleted or incorrectly assumed to have a promotion. The "**Unknown**" category preserves the original uncertainty while allowing the Promotion column to be used consistently in analysis and dashboard filters.

3.4 Business Rules

The following business rules were applied during validation:

- **Total Price must be correct** based on the relevant sales calculation.
- **Discount must not be negative.**
- **Quantity must be greater than 0.**
- Values violating these rules were investigated and corrected or handled appropriately.

3.5 Validation

After cleaning, the dataset was checked for:

- Duplicate records
- Missing values
- Incorrect data types
- Inconsistent categories
- Invalid numerical values
- Outliers
- Business rule violations
- Incorrect return values

Return Value: 1 = Returned and 0 = Not Returned.

4. Analysis & Pivot Tables

Pivot Tables were created from the cleaned dataset to summarize the data and support interactive dashboard visualizations.

Sales Analysis

- Regional Sales Performance
- Customer Type-wise Sales
- Payment Method-wise Sales

- Monthly Sales Trend
- Total Sales, Orders, and Quantity
- Average Discount

Product & Discount Analysis

- Region-wise Product Performance
- Retail vs Wholesale Product Performance
- Discounted vs Non-Discounted Orders
- Quantity Sold by Product
- Average Discount by Customer Type
- Discount by Product

Returns & Order Behavior Analysis

- Returned vs Non-Returned Orders
- Region-wise Return Rate
- Promotion-based Order Performance
- Product-wise Return Rate

These Pivot Tables were used as the **source for KPI cards, charts, slicers, and interactive dashboard analysis.**

5. Dashboard Features

The dashboard provides an interactive view of sales performance for **2023–2025**.

Sales Analysis Dashboard

- Total Sales: ₹16,97,870
- Total Orders: 579
- Total Quantity: 6,054
- Return Rate: 25%
- Average Discount: 0.072
- Regional Sales comparison

- Customer Type Sales analysis
- Payment Method analysis
- Monthly Sales trend

Product & Discount Analysis Dashboard

- Region-wise Product Performance
- Retail vs Wholesale Product Performance
- Discounted vs Non-Discounted Order Analysis
- Quantity Sold by Product
- Average Discount by Customer Type
- Discount by Product

Salesperson & Store Analysis Dashboard

- Orders Completed by salesperson
- Avg Delivery Time
- Sales Performance
- Store Performance

Returns & Order Behavior Dashboard

- Return Status analysis
- Region-wise Return Rate
- Promotion-based Order Performance
- Product-wise Return Rate

Interactive Features

- Timeline: Filter the dashboard by year/date from 2023–2025.
- Product Slicer: Analyze individual products.
- Region Slicer: Compare different regions.
- Dynamic Filtering: Charts and KPI cards update based on selections.
- Multiple Dashboard Views: Allows sales, product/discount, and return behavior to be analyzed separately.

Dashboard Usage: Use the timeline and slicers to dynamically explore sales performance by year, region, product, and customer-related dimensions.....add one more dashboard view at 3ed as salesperson and store analysis

6. Key Insights

Based on the dashboard analysis:

- **Total sales reached ₹16,97,870** across **579 orders**, with a total quantity of **6,054 units**.
 - The overall **return rate is 25%**, meaning approximately one-fourth of the orders were returned.
 - **West region shows the strongest regional sales performance**, while the other regions show relatively close performance.
 - **Laptop and Tablet** show strong product performance across the product-related analysis.
 - The dashboard shows a clear difference between **Retail and Wholesale** customer performance in terms of sales and discounts.
 - **Retail customers have a higher average discount than Wholesale customers** based on the customer-type discount analysis.
 - Discount analysis shows that both **discounted and non-discounted orders** contribute to overall sales.
 - The **Promotion column contains an "Unknown" category** because 370 promotion values were blank and the actual promotion status could not be reliably inferred.
 - Return rates vary across **regions and products**, helping identify areas that may require further investigation.
 - The monthly sales chart shows **variation in sales performance throughout the year**, rather than a constant sales pattern.
 - The dashboard allows these insights to be explored dynamically using **year, region, and product filters**.
-

7. Tools Used

Microsoft Excel

Used for:

- Data analysis
- Pivot Tables
- Pivot Charts
- KPI calculations
- Slicers
- Timeline
- Interactive dashboard

Power Query

Used for:

- Data cleaning
 - Data transformation
 - Handling missing values
 - Standardizing data
 - Data type conversion
 - Validation and preparation of the dataset
-

8. Project Workflow

Raw Data



Data Cleaning & Transformation



Data Validation



Pivot Table Analysis



KPI Calculations



Interactive Dashboard



Business Insights

9. Workbook Structure

The complete project is maintained in a **single Excel workbook** containing:

Sales_Performance_Analysis_2023_2025.xlsx

├── Raw Data

├── Cleaned Data

├── Pivot Tables

└── Dashboard

The **Raw Data** sheet contains the original dataset, while the remaining sheets contain the cleaned data, analysis, supporting calculations, and final interactive dashboard.